

Differentiable Model Compression via Pseudo Quantization Noise

Alexandre Défossez*

Meta AI, FAIR Team, [Paris, France](#)

defossez@fb.com

Yossi Adi*

Meta AI, FAIR Team, [Tel-Aviv, Israel](#)

adiyoss@fb.com

Gabriel Synnaeve

Meta AI, FAIR Team, [Paris, France](#)

gab@fb.com

Reviewed on OpenReview: <https://openreview.net/forum?id=DijnKziche>

Abstract

We propose DIFFQ a differentiable method for model compression for quantizing model parameters without gradient approximations (e.g., *Straight Through Estimator*). We suggest adding independent pseudo quantization noise to model parameters during training to approximate the effect of a quantization operator. DIFFQ is differentiable both with respect to the unquantized weights and the number of bits used. Given a single hyper-parameter balancing between the quantized model size and accuracy, DIFFQ optimizes the number of bits used per individual weight or groups of weights, in end-to-end training. We experimentally verify that our method is competitive with STE based quantization techniques on several benchmarks and architectures for image classification, language modeling, and audio source separation. For instance, on the ImageNet dataset, DIFFQ compresses a 12 layers transformer-based model by more than a factor of 8, (lower than 4 bits precision per weight on average), with a loss of 0.3% in model accuracy. Code is available at github.com/facebookresearch/diffq.

1 Introduction

An important factor in the adoption of a deep learning model for real-world applications is how easily it can be pushed to remote devices. It has been observed that larger models usually lead to better performance, for instance with larger ResNets (He et al., 2016) achieving higher accuracies than smaller ones. In response, the community has worked toward smaller, and more efficient models (Tan & Le, 2019). Yet an EfficientNet-B3 is still almost 50MB, a considerable amount if the model is to be included in online applications, or updated with limited network capabilities. For other applications, such as language modeling (Vaswani et al., 2017) or source separation (Défossez et al., 2019), the typical model size is closer to 1GB, ruling out any kind of mobile usage. Efficient model compression is thus important for on device adoption of deep learning models. Thus, we focus in the present work on reducing model size, rather than achieving computational gains.

The simplest method to reduce model size consists in decreasing the number of bits used to encode individual weights. For instance, using 16 bits floating point numbers halves the model size, while retaining a sufficient approximation of the set of real numbers, $\mathbb{R}$, to train with first-order optimization methods (Micikevicius et al., 2018). When considering lower precision, for instance, 8 or 4 bits, the set of possible values is no longer a good approximation of $\mathbb{R}$, hence preventing the use of first-order optimization methods. Specifically, uniform quantization requires using the `round` function, which has zero gradients wherever it is differentiable.

Quantization can be done as a post-processing step to regular training. However, errors accumulate in a multiplicative fashion across layers, with a possibly uncontrolled decrease in the model accuracy. Courbariaux et al.

*Equal contribution.